

Lab 8 – SVM

DS3010: Introduction to Machine Learning Lab

Timing: 02:00 PM to 04:45 PM

Max Marks:10

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name]_[rollno]_[lab].ipynb
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Write justifications for your choices where needed.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. **Use of generative AI tools (such as ChatGPT, Gemini, etc.) is strictly prohibited. Any submission found to contain AI-generated or plagiarized content will receive a score of zero, and disciplinary action.**

The following code can be used to load mnist dataset

You can use plt.imshow() to display images

```
from sklearn.datasets import fetch_openml

mnist = fetch_openml('mnist_784', version=1, as_frame=False)

X = mnist.data
y = mnist.target.astype(int)
```

```
plt.imshow(X_train[0].reshape(28, 28), cmap="gray")
```

Support Vector Machines on MNIST

Dataset: MNIST handwritten digit dataset

Objective: To understand the behavior of Support Vector Machines (SVMs) by investigating different kernels, regularization parameters, and kernel parameters.

Instructions

- Use the MNIST dataset for all experiments.
- Normalize the pixel values appropriately before training.
- Use the same train-test split for all experiments.
- Record training time, prediction time, training accuracy, and test accuracy.
- Wherever applicable, visualize the results using suitable plots.
- Clearly explain the observations obtained from each experiment.
- Take a subset of images 8000 train and 4000 test for faster results.

Question 1: Linear SVM vs RBF SVM

Train and evaluate two SVM classifiers on the MNIST dataset:

1. A Linear SVM.
2. An SVM with an RBF (Radial Basis Function) kernel.

Keep the other hyperparameters fixed wherever possible.

For both models, report:

- Training accuracy
- Test accuracy
- Training time
- Prediction time

Also generate and compare the confusion matrices of the two models.

Question 2: Effect of the Regularization Parameter C

Using an RBF SVM, investigate the effect of the regularization parameter C .

Use:

$$C \in \{0.01, 0.1, 1, 10, 100\}$$

Keep the kernel and γ fixed.

For each value of C , record:

- Training accuracy
- Test accuracy
- Training time
- Number of support vectors

Note: Try using `plt.semilogx()` for better plots.

Plot:

1. C vs. training accuracy
2. C vs. test accuracy
3. C vs. number of support vectors

Question 3: Effect of the RBF Parameter γ

Using an RBF SVM, investigate the effect of the kernel parameter γ .

Use:

$$\gamma \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$$

Keep $C = 1$ fixed.

For each value of γ , record:

- Training accuracy
- Test accuracy
- Training time
- Number of support vectors

Plot:

1. γ vs. training accuracy
2. γ vs. test accuracy
3. γ vs. number of support vectors

Bonus: Try this without normalizing if you want.

Question 4: SVM vs. Logistic Regression

Using the results from the previous Logistic Regression lab, compare the performance of Logistic Regression with the SVM model from this lab.

Compare their confusion matrices and accuracies.

Question 5: Visualizing Support Vectors

Train an RBF SVM on the MNIST training set and extract the support vectors using the trained model.

Visualize a suitable number of support vectors as 28×28 grayscale images.

Note: If the full MNIST dataset produces too many support vectors to visualize clearly, select a representative subset of support vectors for visualization.

Happy Coding